INCH-POUND

MIL-PRF-1/1159D
17 April 2007
SUPERSEDING
MIL-E-1/1159C
28 March 1967

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 705WA

Inactive for new design
after 30 April 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Diode, high vacuum.

See figure 1.

Mounting position: Any.

Weight: 3.25 ounce nominal.

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	ib	Io	Pp	tk
Unit:	V ac	kv	kv	ma	mA dc	W	sec
Rectifier, half-wave							
(A) Maximum:	5.25	30	---	400	100	60	---
Minimum:	4.75	---	---	---	---	---	20
(B) Maximum:	5.25	15	---	600	150	60	---
Minimum:	4.75	---	---	---	---	---	20
Diode, surge limiting							
Maximum:	5.5	30	10	---	---	75	---
Minimum:	---	---	---	---	---	---	20
TEST CONDITIONS:	5.0	30 (see note 1)	---	400	---	---	20

REQUIREMENT OR TEST:

Qualification: Not required.

Holding period (t = 72 hours).

GENERAL:

Preheating. Prior to testing, all tubes shall be preheated under conditions not exceeding the maximum ratings for sufficient time to obtain stabilized values of electrical characteristics. Preheating may be done in the test set if desired by the manufacturer.

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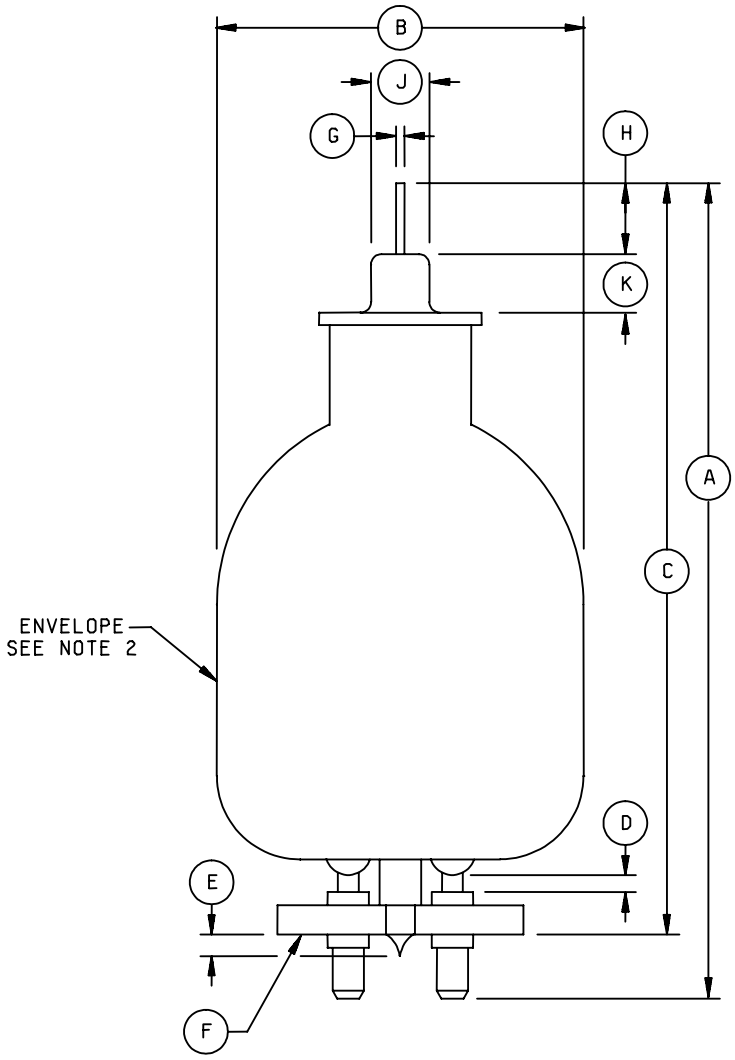

Pin connections	
Pin no.	Element
1	Filament
2	Filament center tap
3	Filament
4	No connection
Cap	Anode

Dimensions (see note 1)				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	4.813	5.062	122.25	128.57
B	---	2.313	---	58.75
C	4.300	4.703	109.22	119.46
Conformance inspection, part 3 (periodic check)				
D	.015	.094	0.38	2.39
E	---	.125	---	3.18
F	Base: A4-52 (see note 3)			
G	.055	.060	1.40	1.52
H	.305	.445	7.75	11.30
J	.368	.382	9.35	9.70
K	.375	---	9.53	---

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based on 1 inch = 25.4 mm.
2. Allowable eccentricity of envelope with respect to centerline through base shall be .126 (3.20 mm) maximum. Conformance inspection, part 3 (periodic check) shall apply.
3. For pin alignment, use gauge GA4-6.

FIGURE 1. Outline drawing of electron tube type 705WA.

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TABLE I. Requirements or tests.

MIL-STD-1311 method or MIL-PRF-1	Requirement or test	Conditions	Acceptance level (see note 8)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part I						
MIL-PRF-1 Appendix D	Visual and mechanical inspection criteria		---	---	---	---	---
1231	Peak emission	eb = 1,000 v (max)	0.65	is	750	---	ma
1256	Anode current	Eb = 300 V dc	0.65	Ib	290	440	mA dc
1301	Filament current	See note 4	0.65	If	4.5	5.25	A
1353	Operation of rectifiers	t = 60 (see notes 5 and 7)	0.65	Io	100	---	mA dc
	Conformance inspection, part 2						
MIL-PRF-1 Appendix D	Dimensions	See figure 1	6.5	---	---	---	---
1101	Secureness of base, base insert, and cap	20 inch-pound torque	6.5	---	---	---	---
1031	High frequency vibration	No voltages	6.5	---	---	---	---
1041	Shock	300 G (see note 6)	6.5	---	---	---	---
---	Post-shock end point	See note 3	6.5	---	---	---	---
	Conformance inspection, part 3						
MIL-PRF-1	Life-test provisions	Group D, operation of rectifiers; T = 125°C; t = 500 hours (see note 2)	---	---	---	---	---
MIL-PRF-1	Life-test end points	Peak emission and failure to rectify.	---	is Io	500 100	--- ---	ma mA dc
	Periodic check tests						
MIL-PRF-1 Appendix D	Dimensions	See figure 1	---	---	---	---	---
1031	Vibration fatigue test	Ef = 5.0 V ac	---	---	---	---	---
---	Post-vibration fatigue test end point	See note 3.	---	---	---	---	---

NOTES:

1. Apply 25 kv epx, then raise epx to 30 kv.
2. Use of heat-dissipating connectors is not permitted during life testing.

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TABLE I. Requirements and tests - Continued.

NOTES - Continued.

3. The end point for vibration fatigue and shock shall be the anode current test (see MIL-STD-1311, method 1256).
4. With 2.5 V ac applied to each half of the filament, each If value shall be within the limits specified herein.
5. No sparking or other irregular operation shall occur during the last 30 seconds of the 1-minute test.
6. Shock-test mounting fixture, as specified on Drawing 288-JAN, or equivalent, shall be used for clamping tube base.
7. This test to be performed at the conclusion of the holding period.
8. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311 and Drawing 288-JAN.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-015)

Review activities:
Army - CR4
Navy - AS, CG, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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